```
praveen@PRAVEENKUMAR:~$ mkdir -p ~/linux-practice/permissions/docs
praveen@PRAVEENKUMAR:~$ cd ~/linux-practice/permissions
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ touch script.sh file.txt
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ touch docs/file1.txt docs/file2.txt
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ echo '#!/bin/bash' > script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ echo 'echo "Hello from WSL"' >> script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ echo 'Sample file' > file.txt
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l
total 12
drwxr-xr-x 2 praveen praveen 4096 Oct 27 10:45 docs
-rw-r--r-- 1 praveen praveen   12 Oct 27 10:45 file.txt
-rw-r--r-- 1 praveen praveen   34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l docs/
total 0
-rw-r--r-- 1 praveen praveen 0 Oct 27 10:45 file1.txt
-rw-r--r-- 1 praveen praveen 0 Oct 27 10:45 file2.txt
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l script.sh
-rw-r--r-- 1 praveen praveen 34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod 755 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l script.sh
-rwxr-xr-x 1 praveen praveen 34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod u+x script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l script.sh
-rwxr-xr-x 1 praveen praveen 34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod -R 644 docs/
chmod: cannot access 'docs/file2.txt': Permission denied
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod 644 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l script.sh
-rw-r--r-- 1 praveen praveen 34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod u+x script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l script.sh
-rwxr--r-- 1 praveen praveen 34 Oct 27 10:45 script.sh
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ chmod -R 644 docs/
chmod: cannot access 'docs/file2.txt': Permission denied
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -l file.txt
-rw-r--r-- 1 praveen praveen 12 Oct 27 10:45 file.txt
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ id alice
id: 'alice': no such user
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo useradd -m alice
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:

praveen@PRAVEENKUMAR:~/linux-practice/permissions$ mkdir -p project
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo groupadd devs
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praveen@PRAVEENKUMAR:~/linux-practice/permissions$ id alice
id: 'alice': no such user
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo useradd -m alice
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:

praveen@PRAVEENKUMAR:~/linux-practice/permissions$ mkdir -p project
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo groupadd devs
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ getent group devs
devs:x:1003:
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo chgrp devs project/
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ ls -ld project/
drwxr-xr-x 2 praveen devs 4096 Oct 27 10:47 project/
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ sudo chgrp -R devs project/
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ stat file.txt
  File: file.txt
  Size: 12              Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 44950       Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ praveen)   Gid: ( 1000/ praveen)
Access: 2026-10-27 10:45:02.525422637 +0000
Modify: 2026-10-27 10:45:23.949410026 +0000
Change: 2026-10-27 10:45:23.949410026 +0000
 Birth: 2026-10-27 10:45:02.525422637 +0000
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ id
uid=1000(praveen) gid=1000(praveen) groups=1000(praveen),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
praveen@PRAVEENKUMAR:~/linux-practice/permissions$ groups praveen
praveen : praveen adm cdrom sudo dip plugdev users
praveen@PRAVEENKUMAR:~/linux-practice/permissions$
```